

PRATYUSH ACHARYA

Data Scientist | NLP Researcher | Applied AI Engineer

Kathmandu, Nepal | +977-9761838106 | acharya.pratyush12@gmail.com
linkedin.com/in/pratyush-acharya | github.com/pratyush-acharya

Profile

Applied NLP researcher and data scientist building reliable retrieval, evaluation, and agentic systems from prototype to production. EMNLP 2026 main-conference author with experience in representation analysis, domain-tuned embeddings, graph RAG, LLM evaluation, and high-throughput document intelligence.

Selected Research

Concepts Whisper: Spectral Anti-Concentration and the Dual Geometry of Transformer Representations

Pratyush Acharya, Nuraj Rimal, Habish Dhakal

Accepted - EMNLP 2026 Main Conference

- Establishes that semantic concept directions systematically occupy spectrally quiet subspaces across 17 transformer models and 22 concept categories, with convergent evidence from difference-of-means vectors, sparse autoencoders, linear probes, and causal interventions.

Experience

SecurityPal AI

Data Scientist

Kathmandu, Nepal

Sep 2024 - Present

- Fine-tuned a domain sentence-transformer on 19,000 labeled question pairs using TripletLoss and a learned pairwise classifier, improving semantic-search accuracy by **21%** while cutting query latency by **3x**; converted similarity graphs into analyst-ready clusters with Leiden and KeyBERT.
- Built and evaluated a Graph-R1 knowledge-graph RAG pipeline using BGE embeddings, ChromaDB, and Gemini; graph traversal produced **63 correct-answer** and **71 no-hallucination** judgments, establishing a product path beyond single-shot vector search.
- Created a reusable customer-readiness evaluation harness that runs **244 common security questions** against each knowledge library and diagnoses content, retrieval, generation, and template failures with question-level recommendations.
- Operationalized a Gemini-powered extraction pipeline for DOC, DOCX, PDF, XLS, and XLSX questionnaires, delivering **30-50x speedups**, processing thousands of questions weekly, and saving an estimated **40-60 analyst-hours per week**.
- Designed modular tool-calling agents that query APIs and databases, reason over structured data, and automate complex operational workflows, reducing end-to-end processing from hours to minutes.
- Developed AnswerPilot and Compliance Lens browser workflows for security-questionnaire QA, including a shipped integrity check that catches inconsistent analyst comments and saves **5-10 minutes per analyst** during ship-outs.

Invisible Technologies

Advanced AI Trainer - Software Development

Kathmandu, Nepal

Apr 2024 - Aug 2024

- Built validation pipelines and manually evaluated LLM-generated code for syntactic correctness, semantic coherence, and compliance with client-specific quality rubrics.
- Curated supervised fine-tuning data and performed high-precision annotation to improve reliability and domain adaptation in production LLM deployments.

Waterflow Technology

Associate AI Engineer

Kathmandu, Nepal

May 2023 - Nov 2023

- Engineered a LangChain and OpenAI API RAG pipeline for document question answering and benchmarked open-source LLMs against GPT-series models to guide model selection.
- Built an Airflow-orchestrated PySpark and Hive ETL pipeline that integrated heterogeneous sources for daily ingestion across multiple business units.

Additional Research

Assessing the Pedagogical Readiness of Large Language Models as AI Tutors in Low-Resource Contexts: A Case Study of Nepal’s K-10 Curriculum

Pratyush Acharya et al.

arXiv:2604.09619

- Introduces a curriculum-aligned benchmark and seven-metric evaluation framework for LLM tutoring, identifying failure modes including the “Expert’s Curse” and “Contextual Blindspot” in Nepalese K-10 education.

Grokking as a Variance-Limited Phase Transition: Spectral Gating and the Epsilon-Stability Threshold

Pratyush Acharya, Habish Dhakal

arXiv:2603.15492

- Models grokking as a variance-gated transition in AdamW optimization, proposing spectral gating and three complexity regimes that govern the shift from memorization to generalization.

Teaching and Mentoring

Deerwalk Training Center

Trainer

Kathmandu, Nepal

Mar 2023 - Present

- Teach data science and machine learning through hands-on projects; designed a beginner-friendly curriculum that prepares learners for advanced AI study and industry practice.

Deerwalk Mentorship Program

Mentor

Kathmandu, Nepal

Mar 2023 - Apr 2025

- Mentored participants in object-oriented programming and real-world implementation using modern frameworks and industry-standard design patterns.

Education

Tribhuvan University

B.Sc. Computer Science and Information Technology

Kathmandu, Nepal

2024

Technical Skills

Languages	Python, SQL
ML and NLP	PyTorch, Hugging Face Transformers, SentenceTransformers, TRL, Scikit-learn, DSPy, LangChain
LLM Systems	RAG and Graph RAG, tool-calling agents, evaluation harnesses, LoRA/QLoRA, prompt optimization, ChromaDB
Data and Infrastructure	BigQuery, PostgreSQL, PySpark, Apache Airflow, Hive, MySQL, MongoDB, Docker, Git, Linux
Research Methods	Transformer interpretability, sparse autoencoders, linear probing, causal intervention, spectral analysis